Theoretically Principled Deep RL Acceleration via Nearest Neighbor Function Approximation

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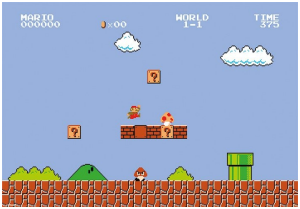
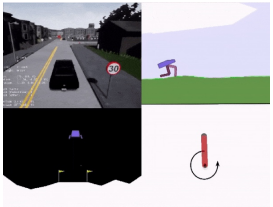
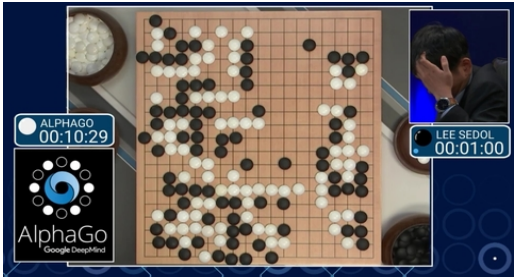
Challenges in Deep RL





- Theoretical foundation of deep RL has not been fully established.
- Model-free deep RL requires considerable samples to learn.
- Online learning coupled with neural networks can be highly unstable.
- Hyperparameter tuning affects learning outcome.

Challenges in RL Theory

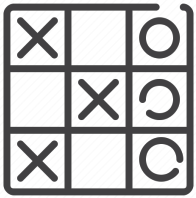


# of states in Go: $3^{19\times 19}$

- Many works focus on tabular RL with small and finite state-action spaces.
- Sample complexity scales linearly on the number of states $|S|$ $\rightarrow$ hard to apply to real-life scenarios.

Motivations

- Can we bridge the gap between RL theory and practice?
  - Theoretically principled function approximator + deep RL framework
- Can we reduce the sample complexity when learning from high-dimensional data?
  - Leverage structural properties of the visual inputs.



Internal state: $(1, 0, 2, 0, 1, 2, 1, 0, 2)$

Pixel representation dimension: 100 x 100

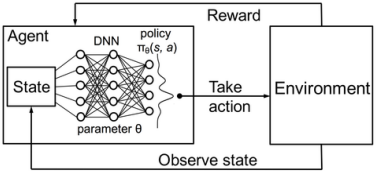
Key idea: human associative learning based on similarity measures. (Treat RL as learning the relationship between state-action pairs and expected cumulative rewards.)

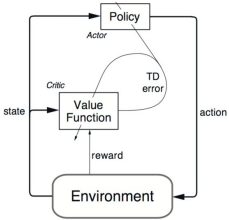
Our Contributions

We do value function approximation using the **nearest neighbor (NN) value function approximator** $\hat{V} : S \times A \to \mathbb{R}$, which maps state-action pairs to expected cumulative rewards.

- Theoretical guarantee: sublinear regret $O\big(H(DLK)^{\frac{d}{d+1}}\big)$ for both low- and high-dimensional inputs in deterministic systems.
- Algorithm 1: Nearest Neighbor Actor-Critic
- Algorithm 2: Nearest Neighbor Update Module

Preliminaries





Deterministic MDP: $(\mathcal{S}, \mathcal{A}, f, r, H)$

Cumulative reward from step h: $R_h = \sum_{t=0}^{H-h} r_{h+t}$
RL objective: $\max_\pi \mathbb{E}_{s_h, a_h \sim \pi}[R_0]$

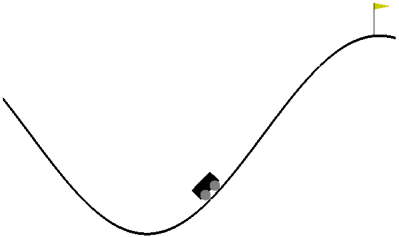
Value function: $V_h^\pi(s) = r(s, \pi(s)) + V_{h+1}^\pi[f(s, \pi(s))]$
Temporal-difference error: $\delta_h = r_h + V_{h+1}^\pi(s_{h+1}) - V_h^\pi(s_h)$
Q-function: $Q_h^\pi(s, a) := r(s, a) + V_{h+1}^\pi\big(f(s, a)\big)$
Bellman optimality equation: $V_h^*(s) = \max_{a \in A} Q_h^*(s, a)$

Sample complexity:
$\text{Regret}(K) = \sum_{k=1}^{K}[V_1^*(s_1^k) - \sum_{h=1}^{H} r(s_h^k, a_h^k)]$

Metric Space and Intrinsic Dimension

Suppose the state-action space $\mathcal{X} = \mathcal{S} \times \mathcal{A}$ is a continuous metric space with a distance function $d_{\mathcal{X}}$ and diameter $D := \sup_{x,x' \in \mathcal{X}} d_{\mathcal{X}}(x, x')$.

- Ambient dimension p: the number of variables *used* to describe a point in $\mathcal{X}$, e.g., the Euclidean dimension of the *natural* embedding of $\mathcal{X}$.
- Intrinsic dimension d: the number of variables *needed* to describe $\mathcal{X}$.



p = 20 x 20 = 400

d = 3 (position, velocity, acceleration)

Value Function Approximation

Given a set of observations $B := \{x_i\}_{i \in [N]}$ and corresponding value labels $\{Q(x_i)\}_{i \in [N]} \subseteq \mathbb{R}$, we want to find a function approximator $f : \mathcal{X} \to \mathbb{R}$ that

1. approximates the known labels with small errors: $f(x_i) \approx Q(x_i)$ for all $i \in [N]$;
2. generalizes to unseen data points: $f(x) \approx Q(x)$ for $x \in \mathcal{X} \backslash B$.

Value Function Approximation

Definition (**Nearest neighbor function approximator**)

Given a sample buffer $B = \{(x_i, Q(x_i))\}_{i \in [N]} \subseteq \mathcal{X} \times \mathbb{R}$*, the nearest neighbor approximator is the function* $\hat{Q} : \mathcal{X} \to \mathbb{R}$ *such that* $\forall x \in \mathcal{X}$*,*

$$\hat{Q}(x) := \min_{i \in [N]} \{Q(x_i) + L \cdot d_{\mathcal{X}}(x, x_i)\}.$$

$L > 0$ *is a parameter that adjusts the approximation error.*

Note: Q matches existing samples exactly and the approximation error for a new x is characterized by an upper bound obtained from the closest known sample.

Assumption 1: Intrinsic MDP satisfies Lipschitz continuity

Assumption (MDP with Lipschitz continuity)

In the metric space $\mathcal{X}$, let the optimal Q-function be $Q_h^* : \mathcal{X} \to \mathbb{R}$, then $\exists L_1, L_2 > 0$ such that $\forall x, x' \in \mathcal{X}, \forall h \in [H]$,

$$|Q_h^*(x) - Q_h^*(x')| \leq L_1 \cdot d_{\mathcal{X}}(x, x'), \tag{1}$$

$$\max_{a''} d_{\mathcal{X}}\left[\left(f(x), a''\right), \left(f(x'), a''\right)\right] \leq L_2 \cdot d_{\mathcal{X}}(x, x'). \tag{2}$$

Theorem: Regret w.r.t. the ambient dimension p

Theorem (Regret till ϵ-optimality by Yang et al. 2019)

Suppose $\mathcal{X}$ has ambient dimension p and admits an ϵ-cover with size $N(\epsilon)$ for any $\epsilon > 0$. After K episodes, UCRL-FA with the nearest neighbor construction obtains a regret bound $\text{Regret}(K) \leq HN(\epsilon) + 2\epsilon LKH$. If $\mathcal{X}$ is compact, then $\text{Regret}(K) = O(DLK)^{\frac{p}{p+1}} \cdot H$.

Assumption 2: Embedding map is Bi-Lipschitz

Suppose the learner observes a p-dimensional state space $\hat{\mathcal{S}}$, e.g., image space, which admits an embedding to a d-dimensional space $\mathcal{S}$, e.g., the parameter space of a physical system, for some $d \ll p$. Let $\mathcal{Y} = \hat{\mathcal{S}} \times \mathcal{A}$ and $d_{\mathcal{Y}}$ be the distance metric satisfying the triangular inequality.

Assumption (Bi-Lipschitz mapping between metric spaces)

There exists a map $g : \mathcal{X} \to \mathcal{Y}$ from the intrinsic state space to the external metric space. We assume that g is Bi-Lipschitz, i.e., $\forall x, x' \in \mathcal{X}$, $\exists C > 0$ such that

$$C^{-1} d_{\mathcal{X}}(x, x') \leq d_{\mathcal{Y}}(g(x), g(x')) \leq C d_{\mathcal{X}}(x, x'). \tag{3}$$

$d_{\mathcal{X}}$ and $d_{\mathcal{Y}}$ are distance metrics for $\mathcal{X}$ and $\mathcal{Y}$, respectively.

Lemma: External MDP satisfies Lipschitz continuity

Lemma (High-dimensional MDP with Lipschitz continuity)

the MDP with state-action space $\mathcal{Y}$ is $\hat{L}$-Lipschitz continuous, where $\hat{L} = (L_2 C^2 + 1) \cdot L_1 C$.

Apply theorem of Yang et al. 2019:

Lemma (Regret in $\mathcal{Y}$ w.r.t. the local dimension)

Suppose that $\mathcal{Y}$ admits an $\hat{\epsilon}$-cover $N(\hat{\epsilon})$ for some $\hat{\epsilon} > 0$, after K episodes, the NN function approximator obtains a regret

$$Regret(K) \leq H \cdot N(\mathcal{Y}, \hat{\epsilon}) + 2\hat{\epsilon}\hat{L}KH \tag{4}$$

where $\hat{L} = (L_2 C^2 + 1) \cdot L_1 C$.

Main theorem: Guarantee for high-dimensional structured input

Theorem (Regret till $\hat{\epsilon}$-optimality for $\mathcal{Y}$ w.r.t. the intrinsic dimension)

Suppose $\mathcal{X}$ admits an ϵ-cover $N(\mathcal{X}, \epsilon)$ and $g : \mathcal{X} \to \mathcal{Y}$, $\mathcal{Y}$ is $\hat{L}$-Lipschitz continuous, then the following holds:

1. *$\mathcal{Y}$ admits an $\hat{\epsilon}$-cover with $\hat{\epsilon} = 2C\epsilon$, $N(\mathcal{Y}, \hat{\epsilon}) \leq N(\mathcal{X}, \epsilon)$*
2. *After K episodes, the NN function approximator obtains a regret bound $O(DMK)^{\frac{d}{d+1}} \cdot H$, where $M := C\hat{L}$.*

Nearest Neighbor Actor-Critic

Algorithm 1 Nearest Neighbor Actor-Critic

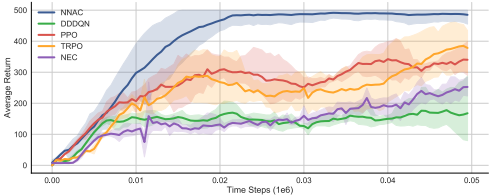
Initialize experience buffer $B^1 = \emptyset$, policy parameter θ_{π^1}
for $k = 1, \ldots, K_{max}$ **do**
  Receive initial state s_1^k
  for $h = 1, \ldots, H$ **do**
    Take action a_h^k according to policy $\pi^k(a_h^k | s_h^k, \theta_{\pi^k})$
    Receive $r_h^k \leftarrow r(s_h^k, a_h^k)$ and $s_{h+1}^k \leftarrow f(s_h^k, a_h^k)$
    $\hat{V}(s_h^k) \leftarrow \text{NNFuncApprox}(s_h^k, h, \pi^k, B^k, H)$
    $\hat{V}(s_{h+1}^k) \leftarrow \text{NNFuncApprox}(s_{h+1}^k, h+1, \pi^k, B^k, H)$
    $\delta_h^k = r(s_h^k, a_h^k) + \gamma \cdot \hat{V}(s_{h+1}^k) - \hat{V}(s_h^k)$
    $B^k \leftarrow B^k \cup \{(s_h^k, a_h^k, f(s_h^k, a_h^k), r(s_h^k, a_h^k), \delta_h^k)\}$
    Sample a random mini-batch of N transitions from B
    Update policy with TD error policy gradient by Eq. (??)
  end for
  $B^{k+1} \leftarrow B^k$
end for

procedure NNFuncApprox(s, h, $\pi(\cdot|\theta_\pi)$, B, H)
  // Assume NN look-up size M
  if $h == H$ **then**
    return 0
  $\mathcal{V} = \emptyset$
  $a \leftarrow \pi(s|\theta_\pi)$
  for $i = 1, \ldots, M$ **do**
    $(s_i, a_i) \leftarrow$ i-th neighbor of (s, a) in B under metric d
    $r_i \leftarrow r(s_i, a_i)$, $s_i' \leftarrow f(s_i, a_i)$ ▷ stored in B
    $\hat{V}_i' \leftarrow \text{NNFuncApprox}(s_i', h+1, \pi(\cdot|\theta_\pi), B, H)$
    $\hat{V}_i \leftarrow r_i + \gamma \cdot \hat{V}_i' + L \cdot d[(s, a), (s_i, a_i)]$
    $\mathcal{V} \leftarrow \mathcal{V} \cup \{\hat{V}_i\}$
  end for
  return $\min \mathcal{V}$

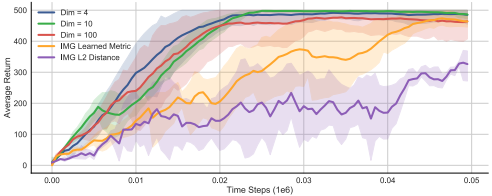
When the NN look-up size is $M = 1$, the **recursive formula** is: $\hat{V}^\pi(x) = r(x) + \gamma \cdot \hat{V}^\pi(x') + L \cdot d_{\mathcal{X}}(x, x')$, where $x' = \arg\min_{x' \in B} d_{\mathcal{X}}(x, x')$ is the nearest neighbor and $\hat{V}^\pi(x') = \hat{Q}^\pi(x', \pi(x'))$.

Nearest Neighbor Actor-Critic

Key idea: policy network + NN value function approximator



(a) NNAC vs baselines (internal state descriptors)



(b) NNAC with different input dimensions

- $\dim(\mathcal{S}) = 4$: 4-tuple descriptors are used directly.
- $\dim(\hat{\mathcal{S}}) = 10$ and 100: we use orthonormal projection matrices to map the 4-tuples into high-dimensional spaces (the distances and the neighbor relations are preserved in the new metric spaces).
- $\dim(\hat{\mathcal{S}}) = 4 \times 20 \times 20$: we use down-sampled gray scale images and stack four consecutive frames together.

Nearest Neighbor Update Module

Algorithm 2 Soft Nearest Neighbor Update

Initialize policy network θ_π, value network θ_Q, $B = \emptyset$, $\alpha \leftarrow \alpha_0$
for each episode **do**
  for each environment step **do**
    Take action a according to $\pi(s)$ and exploration strategy
    Receive reward r and next state s'
    if $\alpha > \epsilon$ **then**
      $\hat{V}(s) \leftarrow \text{NNFuncApprox}(s, h, \pi, B, H)$
      $\hat{V}(s') \leftarrow \text{NNFuncApprox}(s', h+1, \pi, B, H)$
      $\delta = r + \gamma \cdot \hat{V}(s') - \hat{V}(s)$
    end if
    $B \leftarrow B \cup \{(s, a, s', r, \delta)\}$
  end for
  for each gradient step **do**
    Policy gradient: $(1-\alpha) \cdot \nabla_{\theta_\pi} J(\theta_\pi) + \alpha \cdot \delta_{NN} \nabla_{\theta_\pi} \log \pi(a|s; \theta_\pi)$
    Critic loss: $(1-\alpha) \cdot L(\theta_Q) + \alpha \cdot \left\| \delta_{\theta_Q} - \delta_{NN} \right\|^2$
  end for
  $\alpha \leftarrow (1-\beta) \cdot \alpha$
end for

Nearest Neighbor Update Module

- NN value estimation:

$$L'(\theta_Q) = (1-\alpha)\cdot L(\theta_Q) + \alpha \cdot \left\|\delta_{\theta_Q} - \delta_{NN}\right\|^2, \tag{5}$$

- TD-Regularized policy learning:

$$(1-\alpha)\cdot \nabla_{\theta_\pi} J(\theta_\pi) + \alpha \cdot \delta_{NN} \nabla_{\theta_\pi} \log \pi(a|s;\theta_\pi). \tag{6}$$

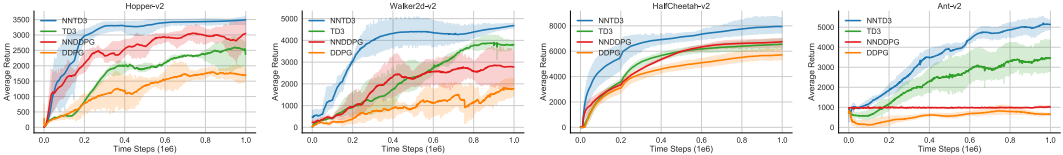


Figure: Learning curves for OpenAI Gym MuJoCo continuous control tasks. Figures from left to right: Hopper-v2, Walker2d-v2, HalfCheetah-v2, and Ant-v2.

Related Work

RL with neural networks: There is a long line of research that applies deep RL to games and control problems (Mnih, Kavukcuoglu, Silver, Graves, et al. 2013; Schulman et al. 2015; Levine et al. 2016; Hasselt, Guez, and Silver 2016; Lillicrap et al. 2016). These results employ several heuristics to accelerate exploration: (Mnih, Kavukcuoglu, Silver, Graves, et al. 2013; Hasselt, Guez, and Silver 2016; Mnih, Kavukcuoglu, Silver, Rusu, et al. 2015) randomly sample actions and store them in a replay buffer before policy learning. Gaussian noise (Lillicrap et al. 2016) or Ornstein-Uhlenbeck noise (Fujimoto et al. 2018) are added to the actions or the network parameters (Plappert et al. 2018) to encourage exploration. Several works also combine model-based value estimation and model-free policy learning to reduce sample complexity (McAllister and Rasmussen 2016; Buckman et al. 2018).

RL with theoretical guarantees: To facilitate theoretical analysis, many works study RL in the tabular setting, where the state and action spaces are discrete (Jaksch, Ortner, and Auer 2008; Azar, Osband, and Munos 2017; Jin, Allen-Zhu, et al. 2018; Zanette and Brunskill 2019). Recently, several works have emphasized understanding in RL with general function approximation (Osband and Roy 2014; Jiang et al. 2017; Sun et al. 2019; R. Wang et al. 2020). However, the function classes are either simple, e.g., linear functions (L. F. Yang and M. Wang 2019; Jin, Z. Yang, et al. 2020), or have strong structural assumptions, which prevent them from being applied to more practical problems.

RL with nearest neighbor search: Model-free episodic control (Blundell et al. 2016) builds on a tabular memory and applies regression using the mean of k-nearest neighbors for Q-value estimation. Neural episodic control (Pritzel et al. 2017) and episodic memory deep Q-networks (Lin et al. 2018) improve the algorithm's generalization ability by absorbing state features into networks. These algorithms take the NN search as a pure classification technique. They do not exploit the fact that the distance between state-action pairs can indicate their relative values. The prototype of our NN function approximator is presented in (L. F. Yang, Ni, and M. Wang 2019), where an upper-confidence algorithm with general function approximation is proposed for tabular RL. The algorithm can apply to continuous cases but requires a discretization of the action space.

Thanks for listening!

GitHub page: https://github.com/sjunhongshen/Accelerating-RL-via-Nearest-Neighbor-Function-Approximation